

1. Administrative & Study Identification

Full Study Title: Phase 2 Study Evaluating Rapcabtagene Autoleucel in Participants With Severe Active GPA or MPA **Short Title/Acronym:** Rapcabtagene Autoleucel in Severe Active GPA/MPA **Protocol Number:** CYTB323I12201 **NCT Number:** NCT06868290 **Sponsor Name / Company:** Novartis / Novartis Pharmaceuticals **Investigational Product:** Rapcabtagene autoleucel **Phase of Development:** Phase II **Trial Status:** Recruiting **Medical Monitor:** Novartis Medical Monitor **Disease Areas:** ANCA Associated Vasculitis (AAV)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of rapcabtagene autoleucel versus comparator in participants with severe active Granulomatosis with Polyangiitis (GPA) or Microscopic Polyangiitis (MPA). **Secondary Objectives:** To evaluate the percentage of patients achieving complete remission, to assess the adjusted annual cumulative glucocorticoid (GC) dose, and to evaluate ANCA seronegativity and sustaining ANCA seronegativity until the analysis cutoff date.

2.2 Background & Rationale Current standard treatments for GPA and MPA often include broad-spectrum immunosuppressants and corticosteroids, which can lead to significant side effects and do not always prevent relapses. Rapcabtagene autoleucel is a CD19-directed CAR T-cell therapy that genetically modifies the patient's T-cells to find and destroy autoreactive B-cells. By eliminating these cells, cell therapy may reset the immune system, offering a precision medicine strategy that could lead to long-term remission with fewer ongoing side effects compared to traditional therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Glucocorticoids) **Blinding:** Open-label, Assessor-blinded **Randomization:** Randomized (Includes a lead-in cohort receiving rapcabtagene autoleucel, followed by a randomized cohort receiving either rapcabtagene autoleucel or comparator)

3.2 Planned Enrollment & Duration Target \$N\$: 126 participants **Number of Sites:** Multinational trial spanning 31 countries

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men and women, aged ≥ 18 and ≤ 75 years with a diagnosis of GPA or MPA according to the American College of Rheumatology/European League Against Rheumatism 2022 (ACR/EULAR 2022) classification criteria. 2. Positive test for ANCA-autoantibodies. 3. GPA and MPA participants with severe active disease.

4.2 Exclusion Criteria 1. Any condition that could prevent a complete washout of medications or could otherwise make the participant ineligible for anti-CD19 CAR-T therapy and further participation in the study. 2. Hypersensitivity and/or contraindications to any product to be given to the participant as part of the study protocol. 3. Other systemic autoimmune diseases requiring therapy. 4. Any medical conditions that are not related to GPA/MPA that would jeopardize the ability of the participant to tolerate CD19 CAR-T cell therapy. 5. Inadequate organ function.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single infusion of rapcabtagene autoleucel. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Single infusion with trial tracking up to approximately 4 years. (Note: After end of study (EOS), participants who received rapcabtagene autoleucel infusion will enter a long-term follow-up (LTFU) period lasting up to 15 years after rapcabtagene autoleucel infusion. This LTFU will be described in a separate study protocol).

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant glucocorticoids as specified per protocol. **Prohibited:** Medications preventing a complete washout prior to CAR-T cell therapy or broad-spectrum immunosuppressants that might interfere with modified T-cell expansion and activity.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Organ Function Assessment, Washout
Baseline	Day 0	Randomization, Cell collection (apheresis), Lymphodepleting conditioning
Interim	Week 13	Efficacy Assessment (Complete Remission Check), Safety Labs
End of Study	Year 4	Final EFS evaluation, GC cumulative dose calculation, transition to 15-year LTFU

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Event-free survival (EFS): Event-free survival (EFS) defined as the time from Randomization to the first occurrence of as per protocol defined events.

Measurement Method: Clinical assessment and tracking of disease-specific adverse events or progression by blinded assessors.

7.2 Secondary & Exploratory Endpoints 1. Percentage of patients achieving complete remission. 2. Adjusted annual cumulative glucocorticoid (GC) dose between Randomization and analysis cutoff date. 3. ANCA seronegativity and sustaining ANCA seronegativity until the analysis cutoff date.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient monitoring for expected CAR T-cell therapy side effects, including fever, tiredness, and low blood cell counts. **Serious Adverse Events (SAEs):** Specific monitoring for cellular therapy-related SAEs such as Cytokine Release Syndrome (CRS) and neurotoxicity, reported within standard 24-hour regulatory timelines. **Data Monitoring Committee (DMC):** External/Internal committee established by Novartis to monitor cohort safety, particularly during the transition from the lead-in to the randomized phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for the randomized cohort evaluating EFS. Safety Set for evaluating CAR-T infusion tolerability. **Sample Size Justification:** $N=126$ designed to achieve adequate statistical power to compare event-free survival differences between the autoleucel arm and the active comparator. **Handling of Missing Data:** Standard survival analysis censoring models (e.g., Kaplan-Meier) for patients lost to follow-up prior to the 4-year limit.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite monitoring post-infusion to manage acute cellular toxicities, followed by periodic remote and onsite follow-ups. **Audit Trail:** Secure eCRF systems capturing blinded evaluations and separate unblinded treatment administration logs.

11. Study Identifiers & Governance

Primary ID: CYTB323I12201 **Registry Number:** NCT06868290 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Phase 2 Study Evaluating Rapcabtagene Autoleucel in Participants With Severe Active GPA or MPA **Official Regulatory Title:** A Phase 2, Randomized, Open-label, Controlled Study to Evaluate the Efficacy and Safety of Rapcabtagene Autoleucel Versus Comparator in Participants With Severe Active Granulomatosis With Polyangiitis (GPA) or Microscopic Polyangiitis (MPA)

12. Clinical Framework (Vasculitis Specific)

Primary Condition: ANCA Associated Vasculitis (AAV), specifically Granulomatosis With Polyangiitis (GPA) and Microscopic Polyangiitis (MPA) **Diagnosis Threshold:** Positive ANCA-autoantibodies and severe active disease presentation **Medical Methodology:** CAR T-cell Therapy (rapcabtagene autoleucel) alongside Glucocorticoids **Optimization Target:** CD19-positive B cell depletion to reset the immune system

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for severe active GPA/MPA compared against experimental CAR-T cell infusion. **Education Protocol (HCP):** Specialized site certification and training for CAR-T administration, CRS management, and long-term immune follow-up. **Education Protocol (Patient):** Onsite training prior to apheresis regarding cell therapy risks, infusion isolation, and long-term symptom monitoring. **Quality Audit Mechanism:** Active tracking of medication washouts, GC dose tapering, and blinded assessor compliance.

14. Observation & Follow-Up Schedule

Total Duration: Up to approx. 4 years in primary protocol (followed by 15-year LTFU) **Onsite Visit Cadence:** Baseline (apheresis/infusion), Week 13, and scheduled safety checkpoints **Remote Monitoring Frequency:** Frequent intervals standard to advanced biological interventions **Checklist Review Interval:** Continuous periodic review of cumulative GC dosage and remission status

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |